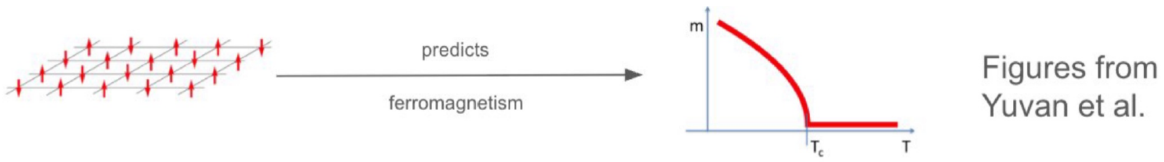
Computational Mechanics

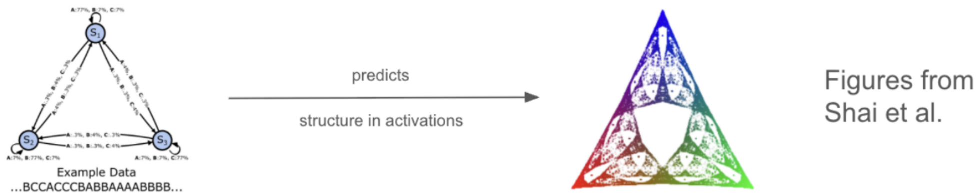
1. Overview & Scope

What is Computational Mechanics?

Statistical Mechanics makes predictions about the behaviour of systems by considering their **statistical** properties. E.g.



Computational Mechanics (comp-mech) makes predictions about the behaviour of systems by considering their **computational** properties. E.g.



Some questions that are central to comp-mech: How much of the past does a system store? How does the system store this information? How does this information affect system behaviour? We will consider each of these questions for transformers today!

Not all systems are interesting! Comp-mech places an emphasis on those performing **optimal prediction**. In this setting the central question becomes:

What are convergent structures relevant for optimal prediction?

Insight on this question greatly constrains our search space when considering **optimal predictors**.

Outline

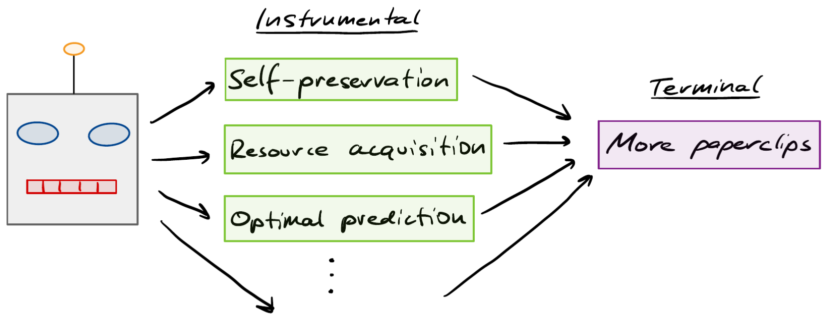
The first part covers what computational mechanics is, computational mechanics and AI safety, where the program is at, and what we will tackle today.

Source: original Part 1 slides, 1–5. Figure credits in the original: Yuvan et al.; Shai et al.. Wording and research status refer to Spring 2026.

Computational Mechanics & AI Safety

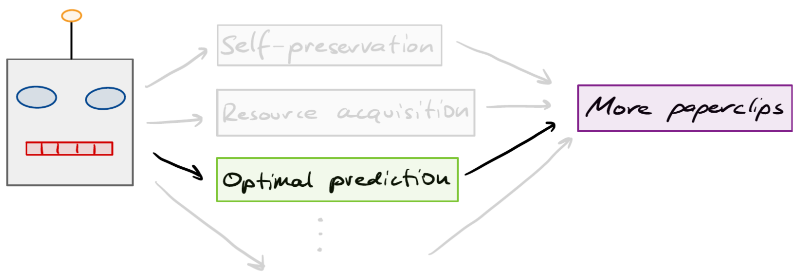
Instrumental and terminal goals

We have strong reason to believe that AI systems will pursue a range of **instrumental goals** in service of their **terminal goal**.



Optimal prediction as an intervention target

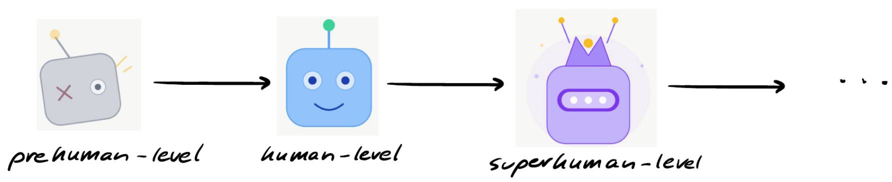
Comp-mech uses an **instrumental goal** as an intervention target point.



By considering internal representations consistent with optimal prediction the search space is reduced.

Capability scale

Targeting an instrumental goal is **scale-free**: it will be relevant to current and future models.



This contrasts with **scale-sensitive** approaches e.g. AI control that are only applicable up to a fixed capability scale. Moreover, as capabilities increase models will better approximate **optimal predictors**.

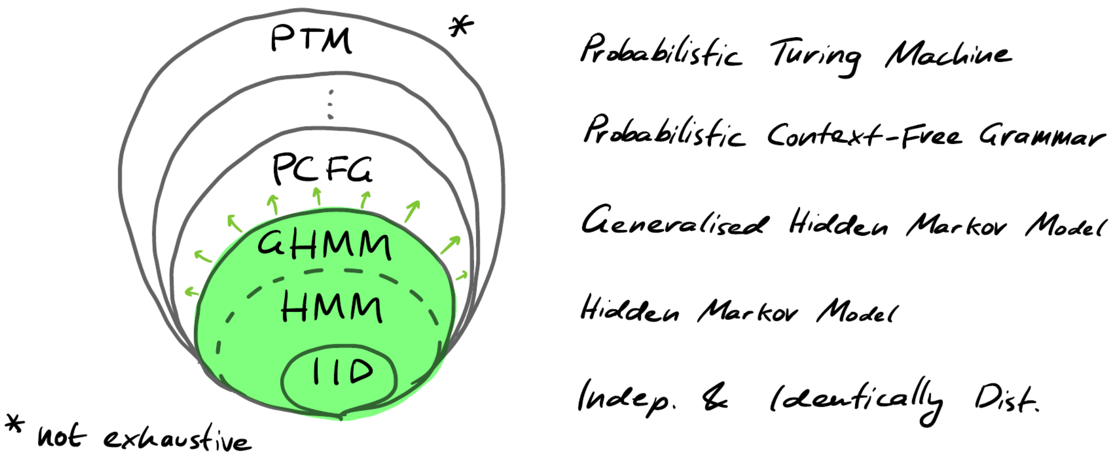
Source: original Part 1 slides, 6–8. Wording and research status refer to Spring 2026.

Overview: where is the program at?

Comp-mech applied to AI safety is a young research program (∼2 years). There is much to do! Progress can be broken down into two orthogonal directions.

Processes

For what classes of data generating **processes** do we understand optimal prediction? Fix a (computable) stochastic process and consider: what resources are required to generate this process? This naturally leads to a hierarchy:



The source diagram marks the region of understood prediction in green and labels the hierarchy as non-exhaustive.

Neural networks

For what scale of **neural networks** have we identified the structures of optimal prediction?

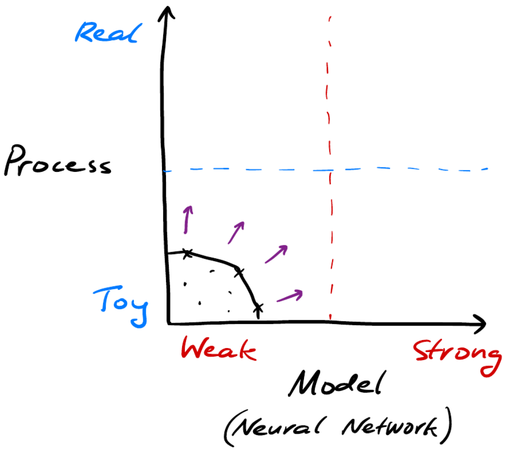
Architecture	Parameters (approximately)
Transformer	10^5
LSTM	10^5
RNN	10^4
GRU	10^5

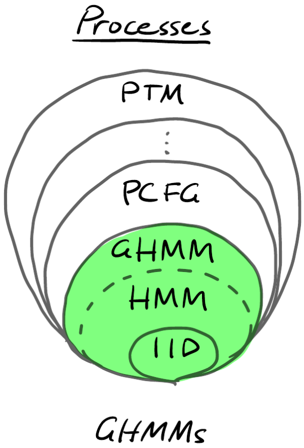
Very toy compared to current frontier models ($\sim 10^{12}$)!

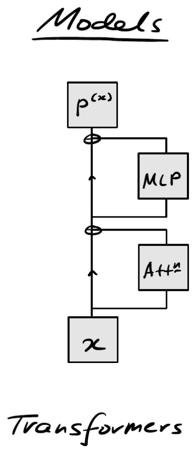
Source: original Part 1 slides, 9–12. Wording and research status refer to Spring 2026.

The research frontier and today's scope

We are working to push the Pareto frontier. We will take a slightly narrower view and consider:







Wild Speculation

As the research program is scale-free one view is that we should develop the foundations such that future automated AI Safety researchers can pursue these ideas independently.

Plan

1	Foundations: Hidden Markov Models	Lecture/exercises
2	Foundations: Prediction	Lecture/exercises
3	Applications: Designing Processes	Tutorial
4	Applications: Belief geometry in transformers	Reading
5	Applications: How do transformers do it?	Tutorial
6	Q & A with Adam Shai and/or Paul Riechers	

Questions?

Source: original Part 1 slides, 13–17. Wording and research status refer to Spring 2026.